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Roll No

EX-8004(2)-CBGS

B.E. VIII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

**Power Electronics Converters for Renewable
Energy**

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1.
 - a) What is need of PWM technique in converter circuits?
 - b) What are different types of PWM technique for multiple pulse forms?
2. Explain single phase and three phase inverters with proper circuit diagram and waveforms.
3.
 - a) What are different types of Renewable Energy Sources? Explain the role of DC-DC converter in variable wind energy conversation system.
 - b) What is the role of Power Electronic Converter's in wind power plants?
4.
 - a) Explain in brief, Photovoltaic Energy Conversion System.
 - b) Write the MPPT Scheme's for solar energy system's and draw their P-V characteristics.

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5. Explain the working of (CSI) Current Source Inverter and state its advantages over (VSI) Voltage Source Inverter.
6. Explain the working of
 - a) STATCON
 - b) SMPS
7. Explain types of battery in detail. Also explain charging and discharging of battery.
8. Explain the following (any two)
 - i) Mitigation of harmonics.
 - ii) Active and Passive filters.
 - iii) Synchronous generator with back to back controlled converter.
 - iv) Doubly fed induction generator with rotor side converter topologies.

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